

Labwork 3: CPU vs GPU Grayscale Conversion

Nguyen Phong Chau - 2540008

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Code

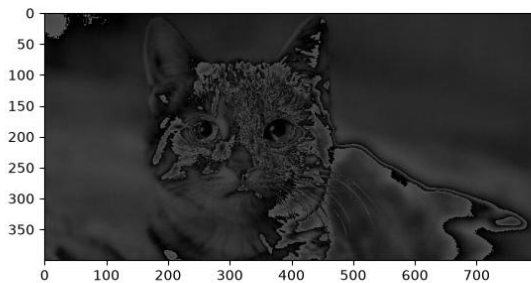
```
blockSizeValues = [32, 64, 128, 256, 512]
img = plt.imread("cat.jpg")
pixelCount = img.shape[0] * img.shape[1]
img_flat = img.reshape(pixelCount, 3)

def grayscale_cpu(src, dst):
    for i in range(src.shape[0]):
        g = np.uint8((src[i, 0] + src[i, 1] + src[i, 2]) / 3)
        dst[i] = g

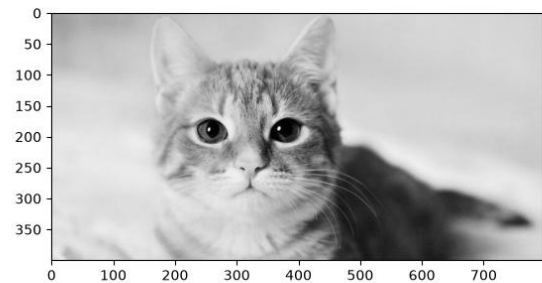
@cuda.jit
def grayscale_gpu(src, dst):
    tid = cuda.threadIdx.x + cuda.blockIdx.x * cuda.blockDim.x
    g = np.uint8((src[tid, 0] + src[tid, 1] + src[tid, 2]) / 3)
    dst[tid, 0] = dst[tid, 1] = dst[tid, 2] = g

devInput = cuda.to_device(img_flat)
devOutput = cuda.device_array((pixelCount, 3), dtype=np.uint8)
grayscale_gpu[gridSize, blockSize](devInput, devOutput)
hostOutput = devOutput.copy_to_host()
```

Block size 64 output



CPU



GPU

Mean time at block size 64 (9 runs): CPU 771.277 ms GPU 1.834 ms

Mean time by block sizes

The performance of GPU is far better than CPU. The block size affects GPU performance slightly, but the GPU remains consistently fast across all tested sizes. For GPU, the best block size is 512, which gives the lowest mean time of 1.329 ms.

Table 1: Average CPU and GPU processing times by block size

Block size	CPU (ms)	GPU (ms)
32	833.801	1.429
64	828.513	1.658
128	837.683	1.653
256	898.968	1.421
512	827.378	1.329

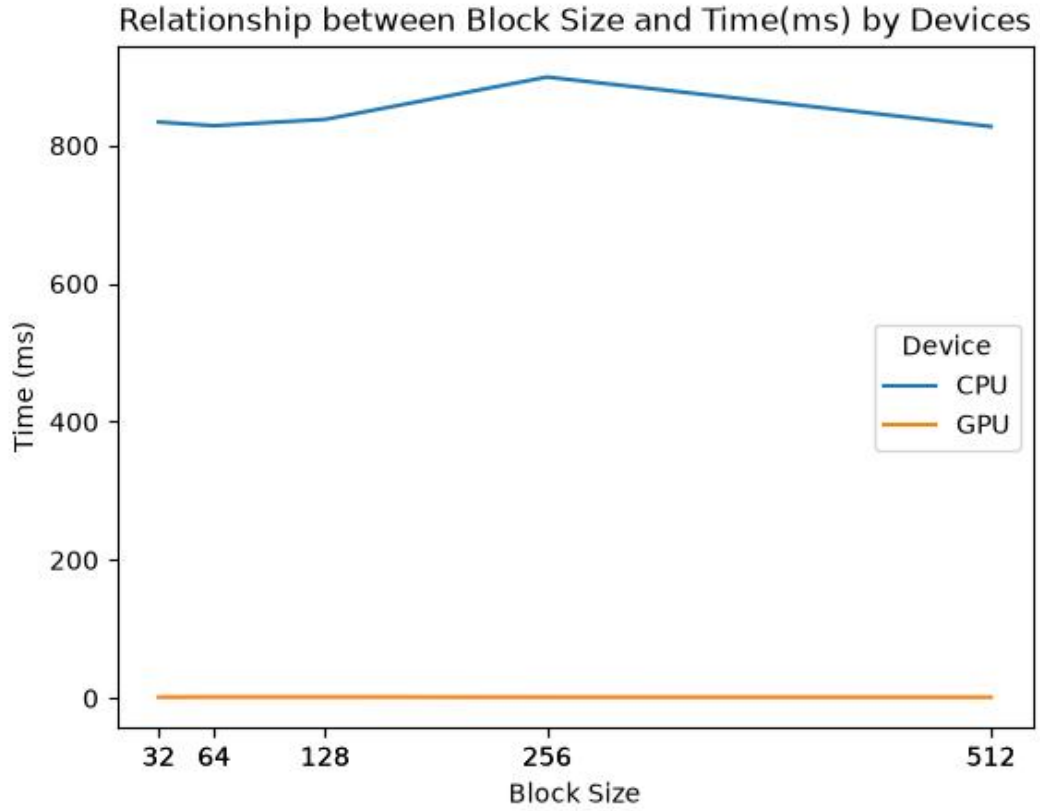


Figure 1: Mean time by block size and device.